SIMATS ENGINEERING

ASSESSMENT TOOL - 1

SCENARIO – BASED QUESTIONS

COURSE NAME: Applied Mathematics

COURSE CODE: UBA06

CO1: Evaluate area as double integrals and apply integral theorems in vector calculus for open and closed surfaces.

Bloom's Levels Covered: L3 – Apply: Assessment Tool Weightage: 10% : Total Marks: 50

Q.No	Question	Bloom's Level	Marks
1	**Hydroelectric Dam Pressure** **Scenario:** A hydraulic engineer needs the net water flux through a closed dam surface. **Questions:** 1. Which theorem determines the total flux? 2. Why is it suitable? 3. State the mathematical theorem.	L3	5
2	**Drone Navigation Wind Field** **Scenario:** An aerospace engineer studies air circulation around a drone's flight path. **Questions:** 1. Which theorem relates circulation to a surface integral? 2. Explain its significance. 3. Mention one aerospace application.	L3	5
3	**Underground Oil Reservoir** **Scenario:** A petroleum engineer estimates the outward flow of oil through a closed reservoir. **Questions:** 1. Which theorem is appropriate? 2. Justify your answer. 3. State the theorem.	L3	5
4	**Smart City Heat Island** **Scenario:** A city planner analyzes the direction of maximum temperature increase. **Questions:** 1. Which vector operator should be used? 2. What does it represent? 3. Give one smart-city application.	L3	5
5	**Offshore Ocean Current** **Scenario:** A marine engineer measures the circulation of ocean currents around a buoy. **Questions:**	L3	5

	1. Identify the suitable theorem. 2. Why is it applicable? 3. State the theorem.		
6	**3D Printing Cooling** **Scenario:** An engineer evaluates heat leaving a closed 3D printed component. **Questions:** 1. Which integral theorem is used? 2. Explain the reason. 3. Mention one manufacturing application.	L3	5
7	**Wind Turbine Blade Rotation** **Scenario:** A renewable energy engineer studies the rotational airflow near turbine blades. **Questions:** 1. Which vector operator measures rotation? 2. What is its physical meaning? 3. Give one renewable energy application.	L3	5
8	**Biomedical Blood Flow** **Scenario:** A biomedical engineer computes blood circulation around an arterial loop. **Questions:** 1. Which theorem converts circulation into a surface integral? 2. Why is it appropriate? 3. Mention one medical application.	L3	5
9	**Satellite Magnetic Mapping** **Scenario:** A space engineer analyzes magnetic circulation around a satellite antenna. **Questions:** 1. Which theorem should be applied? 2. Explain why. 3. State its mathematical form.	L3	5
10	**Industrial Smoke Dispersion** **Scenario:** An environmental engineer determines whether smoke spreads or converges over a factory. **Questions:** 1. Which vector operator should be calculated? 2. What does a positive value indicate? 3. Give one environmental application.	L3	5

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

RUBRIC FOR SCENARIO – BASED QUESTIONS

Criteria	**Excellent (2 Marks)**	**Good (1.5 Marks)**	**Satisfactory (1 Mark)**	**Poor (0.5 Mark)**
Identification of Theorem/Operator	Correctly identifies the appropriate vector operator or integral theorem for the given engineering scenario.	Identifies the correct theorem/operator with minor hesitation or a small error.	Identifies the theorem/operator partially or with noticeable errors.	Incorrectly identifies the theorem/operator or leaves it unanswered.
Conceptual Justification	Provides a clear, logical and accurate justification for selecting the theorem/operator.	Provides a reasonable justification with minor conceptual inaccuracies.	Gives a limited justification with incomplete understanding.	Unable to justify the selection or provides an incorrect explanation.
Mathematical Knowledge	Correctly states the mathematical theorem/formula and explains its significance.	States the theorem/formula correctly with minor omissions or errors.	States the theorem/formula partially or with several mistakes.	Unable to state the theorem/formula correctly.
Engineering Application	Clearly relates the mathematical concept to an appropriate engineering application with relevant examples.	Relates the concept to an engineering application with minor inaccuracies.	Provides a basic or incomplete engineering application.	Unable to relate the concept to any engineering application.
Overall Performance	Demonstrates excellent achievement of CO1 by accurately applying vector calculus concepts to engineering scenarios.	Demonstrates good achievement of CO1 with only minor deficiencies.	Demonstrates satisfactory achievement of CO1 with some conceptual gaps.	Demonstrates inadequate achievement of CO1 learning outcomes.